UCS3461 Foundations of Artificial Intelligence

FIRST-ORDER LOGIC (PREDICATE LOGIC)

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Individuals and Relations

- It is useful to view the world as consisting of individuals (objects, entities) with properties of individuals and relations among individuals.
- Atoms are not symbols. Atoms have an internal structure, a predicate and arguments.
  - S: Socrates is a man.
  - P: Plato is a man.
  - A: Aristotle is a man.
  - H(x): x is a man.
- Reasoning in terms of individuals and relationships can be simpler than reasoning in terms of propositions, if we can express general knowledge that covers all individuals.
- Sometimes you may know some individual exists, but not which one.
- Sometimes there are infinitely many individuals we want to refer to.

Syntax of FOL Formulas

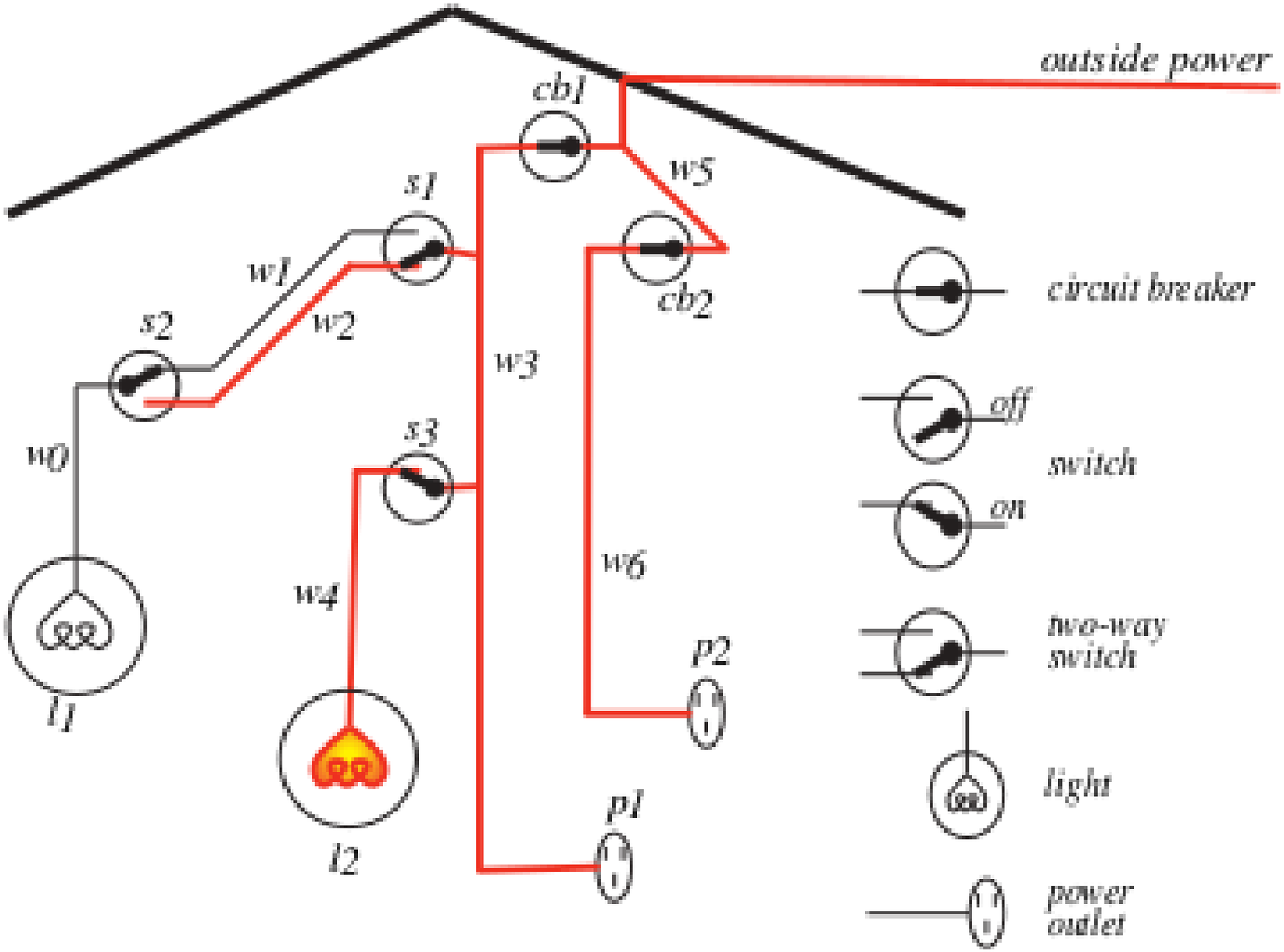
Predicate Logic

- First-Order Logic (FOL)
- Predicate Logic extends propositional logic in two ways:
  - atoms have structure and can include individuals (constants and variables)
  - quantification of variables.

Building Blocks

- A constant denotes an individual in the domain.
  Examples: s2, l2
- A variable is a placeholder for an individual.
- A function symbol denotes a function in the domain.
- A predicate symbol denotes a relation on the domain.
  Examples: connected, lit
- A term is either a variable or a constant or a function.
- An atomic formula (atom) is of the form P or $P(t_1, \ldots, t_n)$ where P is a predicate symbol and t_i are terms.
  Examples: Connected(w0, w1), Lit(l2).
- Logical connectives: $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (if-then), $\leftrightarrow$ (if and only if)
- Quantification: $\forall$ (for all), $\exists$ (there exists)

Example: Electrical Environment



$Light(l_1).$

$Light(l_2).$

$Down(s_1).$

$Up(s_2).$

$Up(s_3).$

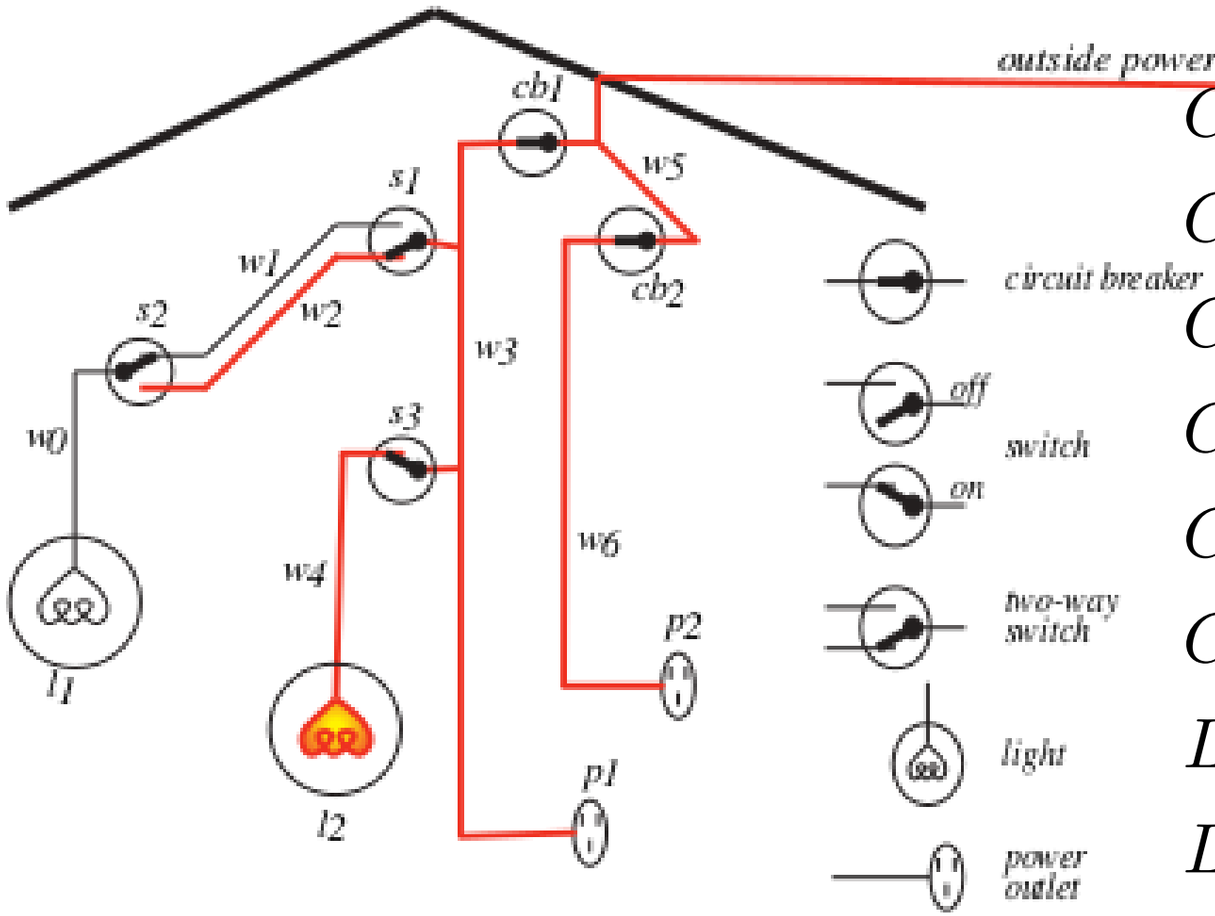
$Ok(l_1).$

$Ok(l_2).$

$Ok(cb_1).$

$Ok(cb_2).$

Example: Electrical Environment



$Connected_to(w_0, w_1) \leftarrow up(s_2).$

$Connected_to(w_0, w_2) \leftarrow down(s_2).$

$Connected_to(w_1, w_3) \leftarrow up(s_1).$

$Connected_to(w_2, w_3) \leftarrow down(s_1).$

$Connected_to(w_4, w_3) \leftarrow up(s_3).$

$Connected_to(p_1, w_3).$

$Lit(L) \leftarrow Light(L) \wedge Ok(L) \wedge Live(L$

$Live(Y) \leftarrow$

$\quad Connected_to(Y, Z) \wedge Live(Z).$

$Live(outside)$

Terms

- A variable is a term.
- A constant symbol is a term.
- If f is a function symbol with arity k, and if $t_1, \ldots, t_k$ are terms, then $f(t_1, \ldots, t_k)$ is a term.
- A constant symbol is a function symbol with 0-arity.

FOL Formulas

- P is a predicate symbol with arity k, and if $t_1, \ldots, t_k$ are terms, then $P(t_1, \ldots, t_k)$ is a formula.
- If A is a formula, $(\neg A)$ is a formula.
- If A and B are formulas, $(A \wedge B), (A \vee B), (A \rightarrow B)$ and $(A \leftrightarrow B)$ are formulas.
- If x is a variable and A is a formula, then $\forall x\ A$ and $\exists x\ A$ are formulas.

Note:

- A predicate symbol with its arguments is an atomic formula (or atom).
- An atomic formula with no occurrence of variables is a ground atom.
- A ground atom has a truth value.
- Predicate symbols with zero arguments are propositions.

Quantifiers

- Assert a relation or a property about all individuals or some individuals in a set.
- The set of individuals about which we want to assert a relation or a property is called the domain.
- Universal quantifier $\forall x\ A$
- Existential quantifier $\exists x\ A$

Universal Quantifier

- Assert that a relation or a property holds for all individuals in the domain.
- $D = \{-1, 0, 1, 2, 4\}$.
- $L(x, y)$: "x is less than y"
- Every individual in D is less than 5 is denoted by

$$\forall x \; L(x, 5)$$

Meaning of Universal Quantifier

- Let the domain $D = \{2, 3, 5, 7\}$ and the predicate formula $P(x)$ denote the relation "x is prime". Then

$$\forall x P(x) \equiv P(2) \wedge P(3) \wedge P(5) \wedge P(7)$$

- Universal quantifier is an abbreviation for conjunction in a finite domain. If the domain is finite $D = \{x_1, x_2, \ldots, x_n\}$, then

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n)$$

Existential Quantifier

- Assert that a relation or a property holds for some individuals in the domain.
- $D = \{1, 2, 4, 6, 7\}$
- $L(x, y)$: "x is less than y"
- Some individuals in D are less than 5 is denoted by

$$\exists x \; L(x, 5)$$

Meaning of Existential Quantifier

- Let the domain $D = \{2, 3, 8, 9\}$ and the predicate formula $P(x)$ denote the relation "x is prime". Then

$$\exists x P(x) \equiv P(2) \vee P(3) \vee P(8) \vee P(9)$$

- Existential quantifier is an abbreviation for disjunction in a finite domain. If the domain is finite $D = \{x_1, x_2, \ldots, x_n\}$, then

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$$

Bound Occurrences and Free Occurrences

- Scope: In $\forall x A(x)$ ($\exists x A(x)$), $A(x)$ is the scope of the quantifier $\forall x$ ($\exists x$).
- Every occurrence of a quantified variable in its scope is a bound occurrence.
- A variable with a bound occurrence is called a bound variable.
- Every occurrence of a variable which is not a bound occurrence is a free occurrence.
- A variable which has only free occurrence is called a free variable.

$\forall x \; A(x)$ ← Scope of $\forall x$

Quantified variable
Bound variable

Bound occurrence

Scope, Bound Occurrences: Example

$\forall x\ Lit(x) \leftarrow Light(x) \wedge Ok(x) \wedge Live(x).$

$\forall y \forall z\ Live(y) \leftarrow Connected_to(y, z) \wedge Live(z).$

$Live(outside).$

$\forall x\ (Lit(x) \leftarrow Light(x) \wedge Ok(x) \wedge Live(x)).$

$\forall y \forall z\ (Live(y) \leftarrow Connected_to(y, z) \wedge Live(z)).$

$Live(outside).$

Quantifiers: Examples

Represent the sentences using the predicate $Loves(x, y)$, which stands for "x loves y" and suitable quantifiers

1. Everybody loves everybody
2. Everybody loves somebody
3. Someone is loved by everyone
4. Somebody loves everybody

Semantics of FOL Formulas

Semantics: General Idea

- A semantics specifies the meaning/value of sentences in the language.
- An interpretation specifies:
  - what individuals (objects) are in the world
  - the correspondence between symbols in the language, and individuals and relations in world
    - predicate symbols denote relations
    - constant symbols denote individuals
    - variables denote individuals
    - function symbols denote functions

Interpretation

- Let A be a formula where $\{P_1, \ldots, P_m\}$ are all the predicate symbols appearing in A and $\{a_1, \ldots, a_k\}$ are all the constant symbols/variables appearing in A. An interpretation I for A is a triple:

$$I = (D, \{R_1, \ldots, R_m\}, \{d_1, \ldots, d_k\})$$

where

  - D is a non-empty set of individuals called the domain,
  - R_i is an n-ary relation on D that is assigned to the n-ary predicate symbool p_i and
  - $d_j \in D$ is a domain element assigned to the constant symbol/variable a_j.

Variable Assignment

- Let I be an interpretation for a formula A. Interpretation $I' = I[x_i/d_i]$ is the same as the interpretation I except that variable x_i is mapped to (replaced by) domain element d_i .

Evaluating a Formula

Let A be a formula, and I an interpretation. The truth value of A under I is defined by induction on the structure of A

- Let $A = P_k(a_1, \ldots, a_n)$ be an atomic formula where each a_i is either a variable x_i or a constant symbol c_i.
  $I(A) = T$ iff $(d_1, \ldots, d_n) \in R_k$ where d_i is the domain element assigned to a_i, and R_k is the relation assigned by I to P_k.
- $I(\neg A) = T$ iff $I(A) = F$.
- $I(A_1 \vee A_2) = T$ iff $I(A_1) = T$ or $I(A_2) = T$, and similarly for the other logical connectives.
- $I(\forall x\ A) = T$ iff $I[x/d](A) = T$ for all $d \in D$.
- $I(\exists x\ A) = T$ iff $I[x/d](A) = T$ for some $d \in D$.

Interpretation: Example

- $\forall x\ P(a, x)$
  - $I = (N, \{\leq\}, \{0\})$

$$\begin{aligned}\forall x\ P(a, x) &\equiv \forall x\ \ \leq (0, x) \\ &\equiv 0 \leq 0 \wedge 0 \leq 1 \wedge 0 \leq 2 \ldots \\ &\equiv T \wedge T \wedge T \ldots \\ &\equiv T\end{aligned}$$

  - $I = (N, \{\leq\}, \{1\})$
  - $I = (Z, \{\leq\}, \{0\})$

Interpretation: Example

- $\forall x\ P(x, f(x)) \wedge Q(g(a, z))$

$$
\begin{aligned}
D &= \{0, 1, 2, \ldots\} \\
I(P) &= \{(m, n) \mid m, n \in D \text{ and } m < n\} \\
I(Q) &= \{n \in D \text{ is prime}\} \\
I(f) &= f(n) = n + 1 \\
I(g) &= g(m, n) = m + n \\
I(a) &= 2 \\
I(z) &= 3
\end{aligned}
$$

Inference Rules in FOL

Equivalences with Quantifiers

1. $$\neg\forall x\ A \equiv \exists x\ \neg A$$ DeMorgan's Law

   $$\neg\exists x\ A \equiv \forall x\ \neg A$$ DeMorgan's Law

2. $$\forall x\ \forall y\ A \equiv \forall y\ \forall x\ A$$ $\forall$ commutative

   $$\exists x\ \exists y\ A \equiv \exists y\ \forall x\ A$$ $\exists$ commutative

3. $$\forall x(A \wedge B) \equiv \forall x A \wedge \forall x B$$

   $$\exists x(A \vee B) \equiv \exists x A \vee \exists x B$$

4. If x does not occur free in B

   $$\forall x\ A \wedge B \equiv \forall x\ (A \wedge B)$$

   $$\forall x\ A \vee B \equiv \forall x\ (A \vee B)$$

   $$\exists x\ A \vee B \equiv \exists x\ (A \vee B)$$

   $$\exists x\ A \wedge B \equiv \exists x\ (A \wedge B)$$

Inference Rules with Quantifiers

$$\frac{\forall x\ A(x)}{A[x/t]}$$ $\forall$ elimination

$$\frac{A(t)}{\forall x\ A(x)}$$ $\forall$ introduction

$$\frac{\exists x\ A(x)}{A[x/t]}$$ $\exists$ elimination

$$\frac{A(t)}{\exists x\ A(x)}$$ $\exists$ introduction

$\forall$ Elimination, $\exists$ Introduction

$\forall$ Elimination

$$\frac{\forall x\ A(x)}{A(t)}$$

m	$\forall x\ A(x)$	
	$\vdots$	
n	$A[x/t]$	$\forall$E, m

$\exists$ Introduction

$$\frac{A(t)}{\exists x\ A(x)}$$

m	$A(t)$	
	$\vdots$	
n	$\exists x\ A(x)$	$\exists$I, m

$\exists$ Elimination

$$\frac{\exists x\ A(x)}{A(t)}$$

p	$\exists x\ A(x)$		
	$\vdots$		
m	t	$A[x/t]$	
		$\vdots$	
n		B	
$n+1$	B		$\exists$E, p, m–n

- t must be a fresh name in the subproof
- t must not occur in B

$\forall$ Introduction

$$\frac{A(t)}{\forall x\ A(x)}$$

m	t	
$\vdots$		$\vdots$
n		$A(t)$
$n+1$	$\forall x\ A(x)$	$\forall$I, m–n

- t must not occur in any live assumption for $A(t)$.
- t must not occur in $\forall x\ A(x)$

Example

- All human beings are mortal.
- Socrates is human.
- Therefore, Socrates is mortal.

$$\forall x(H(x) \rightarrow M(x)), H(s) \therefore M(s)$$

1	$\forall x(H(x) \rightarrow M(x))$	
2	$H(s)$	
3	$H(s) \rightarrow M(s)$	$\forall$ E, 1
4	$M(s)$	$\rightarrow$ E, 2, 3

Example

$$\forall x(H(x) \rightarrow M(x)), \exists x H(x) \therefore \exists x M(x)$$

1	$\forall x(H(x) \rightarrow M(x))$
2	$\exists x H(x)$
	$\vdots$
n	$\exists x M(x)$

Example

1	$\exists x(A(x) \wedge B(x))$
	$\vdots$
n	$\exists x A(x) \wedge \exists x B(x)$

Normal Form

Conversion of a Formula to CNF

- Every formula can be transformed into an equivalent formula in CNF.
- Steps
  1. Eliminate $\leftrightarrow$
     $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$
  2. Eliminate $\rightarrow$
     $P \rightarrow Q \equiv \neg P \vee Q$
  3. "Move $\neg$ inward" to the atoms by applying De Morgan's laws
     $\neg(P \vee Q) \equiv \neg P \wedge \neg Q, \neg(P \wedge Q) \equiv \neg P \vee \neg Q$
  4. Eliminate double negation
     $\neg\neg P \equiv P$
  5. Apply distribuitive laws
     $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R), P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Substitution

- Let A be a formula, x a variable, and t a term. Then, $A[x/t]$ denotes the formula, obtained from A by substituting t for every free occurrence of x in A.
- Read $A[x/t]$ as "A with x replaced by t".

Prenex Form

- A formula of the form $Q_1 y_1 Q_2 y_2 \ldots Q_n y_n \; A$, where Q_i is a quantifier, y_i are variables, and A does not contain a quantifier, is in prenex form.
- $Q_1 y_1 Q_2 y_2 \ldots Q_n y_n$ is the prefix and A is the matrix.

$$\forall x \, (\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz$$
$$\forall x \, ((\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz)$$
$$\forall x \, (\exists y \, (\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz)$$
$$\forall x \exists y \, ((\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz)$$
$$\forall x \exists y \forall w ((\neg Pxz \wedge \neg Qxy) \vee Pwz)$$

Normal Form in FOL

- Standardize apart variables.
- Elimitate arrows and push negation to atoms.
- Extract quantifiers from the matrix and transform it to prenex form.
- Eliminate $\exists$ quantifiers.
- Apply distributive laws and transform it to CNF (and clausal form).

Normal Form

1. Standardize apart/Rectify: systematically rename the bound variables.
2. Eliminate arrows: Eliminate $\leftrightarrow$ and $\rightarrow$ by applying $a \leftrightarrow b \equiv (a \rightarrow b) \wedge (b \rightarrow a)$ and $a \rightarrow b \equiv \neg a \vee b$.
3. Push negation to atoms by applying De Morgan's laws.
4. Prenex form: Convert the formula to prenex form (quantifiers form a prefix).
5. Skolem form: Eliminate the existential quantifiers by converting the formula into Skolem form.
6. CNF: Convert the matrix of the formula into CNF by applying distributive laws (as a set of clauses).

Rectifed Form

- Standardize apart (Rectify) a formula : Systematically rename bound variables so that each bound variable is distinct.
- Rectified form: If no variable occurs both bound and free and if all quantifiers in the formula refer to different variables.
- Rename bound variables: Let $Qx\ B$ be a formula where $Q \in \{\forall, \exists\}$. Then, $Qx\ B \equiv Qy\ B[x/y]$,
  provided y is a variable that does not occur free in B.

Formula	$\neg\exists x\,(Pxz \vee \forall y Qxy) \vee \forall y Pyz$
Rectify	$\equiv \neg\exists x\,(Pxz \vee \forall y Qxy) \vee \forall w Pwz$

Push Negation to Atoms

Rectified	$\neg\exists x\,(Pxz \vee \forall y Qxy) \vee \forall w Pwz$
Negation to atoms	$\equiv \forall x \neg\,(Pxz \vee \forall y Qxy) \vee \forall w Pwz$
	$\equiv \forall x\,(\neg Pxz \wedge \neg\forall y Qxy) \vee \forall w Pwz$
	$\equiv \forall x\,(\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz$

Prenex Form

- A formula of the form $Q_1 y_1 Q_2 y_2 \ldots Q_n y_n \; A$, where Q_i is a quantifier, y_i are variables, and A does not contain a quantifier, is in prenex form.

$$
\begin{aligned}
& \forall x\,(\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz \\
\text{Prenex} \quad & \equiv \forall x\,((\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz) \\
& \equiv \forall x\,(\exists y\,(\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz) \\
& \equiv \forall x \exists y\,((\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz) \\
& \equiv \forall x \exists y \forall w\,((\neg Pxz \wedge \neg Qxy) \vee Pwz)
\end{aligned}
$$

- Choose an outermost quantifier – it is not within the scope of another quantifier.
- Extract the quantifier using the equivalences (Q is a quantifier, op is either $\wedge$ or $\vee$).

$$
\begin{aligned}
Qx\; Ax \text{ op } B & \equiv Qx\;(Ax \text{ op } B) \\
A \text{ op } Qx\; Bx & \equiv Qx\;(A \text{ op } Bx)
\end{aligned}
$$

Skolem Form

While A contains an existential quantifier

- Let A have the form $A = \forall y_1 \forall y_2 \ldots \forall y_n \exists z\ B$ for some formula B in RPF and $n > 0$
- Let f be a new function symbol of arity n that does not yet occur in A;

$$A := \forall y_1 \forall y_2 \ldots \forall y_n\ B[z/f(y_1, y_2, \ldots, y_n)]$$

i.e. the existential quantifier $\exists z$ in A is canceled and each occurence of the variable z in B is substituted by $f(y_1, y_2, \ldots, y_n)$

Skolemization: Example

$$\forall x \exists y \forall w \, ((\neg Pxz \wedge \neg Qxy) \vee Pwz)$$

Skolemize $$\approx \forall x \forall w \, ((\neg Pxz \wedge \neg Qxf(x)) \vee Pwz)$$

CNF and Clausal Form

- Drop $\forall$ quantifiers.
- CNF: Convert the matrix of the formula into CNF (and write down the resulting formula as a set of clauses).

$$\forall x \forall w\, ((\neg Pxz \wedge \neg Qxf(x)) \vee Pwz)$$

Distribute matrix $\equiv (\neg Pxz \vee Pwz) \wedge (\neg Qxf(x) \vee Pwz)$

Clause form $\equiv \{\neg Pxz, Pwz\}, \{\neg Qxf(x), Pwz\}$

Normal Form: Example

$\neg\exists x\,(Pxz \vee \forall y Qxy) \vee \forall y Pyz$	Formula
$\equiv \neg\exists x\,(Pxz \vee \forall y Qxy) \vee \forall w Pwz$	Rectify
$\equiv \forall x\,(\neg Pxz \wedge \neg\forall y Qxy) \vee \forall w Pwz$	Negation to atoms
$\equiv \forall x\,(\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz$	
$\equiv \forall x\,((\neg Pxz \wedge \exists y \neg Qxy) \vee \forall w Pwz)$	Prenex
$\equiv \forall x\,(\exists y\,(\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz)$	
$\equiv \forall x \exists y\,((\neg Pxz \wedge \neg Qxy) \vee \forall w Pwz)$	
$\equiv \forall x \exists y \forall w\,((\neg Pxz \wedge \neg Qxy) \vee Pwz)$	
$\approx \forall x \forall w\,((\neg Pxz \wedge \neg Qxf(x)) \vee Pwz)$	Skolemize
$\equiv (\neg Pxz \vee Pwz) \wedge (\neg Qxf(x) \vee Pwz)$	CNF
$\equiv \{\neg Pxz, Pwz\}, \{\neg Qxf(x), Pwz\}$	Clause form

Normal Form

1. Standardize/Rectify: systematically rename bound variables.
2. Eliminate arrows: Eliminate $\leftrightarrow$ and $\rightarrow$ by applying $a \leftrightarrow b \equiv (a \rightarrow b) \wedge (b \rightarrow a)$ and $a \rightarrow b \equiv \neg a \vee b$.
3. Push negation to atoms by applying De Morgan's laws.
4. Prenex form: Convert the formula to prenex form (quantifiers form a prefix).
5. Skolem form: Eliminate the existential quantifiers by converting the formula into a Skolem form.
6. CNF and Clausal form: Convert the matrix of the formula into CNF by applying distributive laws (as a set of clauses).

Examples

$$\forall x(\forall y A(y) \rightarrow L(x,y)) \rightarrow \exists y\ L(y,x)$$

$$\forall x \exists y \forall z \exists w(\neg P(a,w) \vee Q(x,y))$$

$$\neg \exists x\ (P(x,z) \vee \forall y Q(x,y)) \vee \forall y P(x,y,z)$$

$$\exists x \forall y P(x,y) \rightarrow \forall y \exists x P(x,y)$$

Normalization

Formula	$\exists x \forall y P(x,y) \rightarrow \forall y \exists x P(x,y)$
Standardize variables	$\exists x \forall y P(x,y) \rightarrow \forall w \exists z P(z,w)$
Eliminate arrows	$\neg \exists x \forall y P(x,y) \vee \forall w \exists z P(z,w)$
Push negation to atoms	$\forall x \exists y \neg P(x,y) \vee \forall w \exists z P(z,w)$
Prenex	$\forall x (\exists y \neg P(x,y) \vee \forall w \exists z P(z,w))$
	$\forall x \exists y (\neg P(x,y) \vee \forall w \exists z P(z,w))$
	$\forall x \exists y \forall w (\neg P(x,y) \vee \exists z P(z,w))$
	$\forall x \exists y \forall w \exists z (\neg P(x,y) \vee P(z,w))$
Skolem	$\forall x \forall w \exists z (\neg P(x,f(x)) \vee P(z,w))$
	$\forall x \forall w (\neg P(x,f(x)) \vee P(g(x,w),w))$
CNF	$\neg P(x,f(x)) \vee P(g(x,w),w)$
Clausal form	$\{\{\neg P(x,f(x)), P(g(x,w),w)\}\}.$

Normalization

Formula	$\exists x \forall y Pxy \rightarrow \forall y \exists x Pxy$
Standardize variables	$\equiv \exists x \forall y Pxy \rightarrow \forall w \exists z Pzw$
Eliminate arrows	$\equiv \neg \exists x \forall y Pxy \vee \forall w \exists z Pzw$
Push negation to atoms	$\equiv \forall x \exists y \neg Pxy \vee \forall w \exists z Pzw$
Prenex	$\equiv \forall x (\exists y \neg Pxy \vee \forall w \exists z Pzw)$
	$\equiv \forall x \exists y (\neg Pxy \vee \forall w \exists z Pzw)$
	$\equiv \forall x \exists y \forall w (\neg Pxy \vee \exists z Pzw)$
	$\equiv \forall x \exists y \forall w \exists z (\neg Pxy \vee Pzw)$
Skolem	$\equiv \forall x \forall w \exists z (\neg Pxf(x) \vee Pzw)$
	$\approx \forall x \forall w (\neg Pxf(x) \vee Pg(x,w)w)$
CNF	$\equiv \neg Pxf(x) \vee Pg(x,w)w$
Clausal form	$\equiv \{\{\neg Pxf(x), Pg(x,w)w\}\}.$

End